

Question 1: Create reusable functions for modular code by performing the following 3 tasks:

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Total cost: \$24.18

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Original string: capitalizes the first letter of each word

Title case string: Capitalizes The First Letter Of Each Word

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Discounted price: \$9.89

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Question 2: Manipulate and manage arrays using built-in PHP functions by performing the following 3 tasks:

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Before removing duplicates:

Apple: \$0.30  
Banana: \$0.20  
Orange: \$0.10  
Pineapple: \$0.50  
Grape: \$0.40  
Strawberry: \$0.60  
Watermelon: \$2.00  
Mango: \$1.50  
Kiwi: \$0.80  
Apple: \$0.30  
Banana: \$0.20  
Orange: \$0.10  
Pineapple: \$0.50

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After removing duplicates and sorting by price:

Orange: \$0.10  
Banana: \$0.20  
Apple: \$0.30  
Grape: \$0.40  
Pineapple: \$0.50  
Strawberry: \$0.60  
Kiwi: \$0.80  
Mango: \$1.50  
Watermelon: \$2.00

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Original product list:

Laptop (Electronics): \$999.99  
Smartphone (Electronics): \$699.99  
Headphones (Electronics): \$149.99  
Tablet (Electronics): \$399.99  
Smartwatch (Electronics): \$299.99  
Camera (Electronics): \$549.99  
Apple (Fruit): \$0.30  
Banana (Fruit): \$0.20  
Orange (Fruit): \$0.10  
Grape (Fruit): \$0.40  
Book (Education): \$15.99  
Notebook (Education): \$4.99  
Pen (Education): \$1.99  
T-Shirt (Clothing): \$19.99

Jeans (Clothing): \$49.99  
Sneakers (Clothing): \$79.99

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After applying 10% discount to Electronics products:

Laptop (Electronics): \$899.99  
Smartphone (Electronics): \$629.99  
Headphones (Electronics): \$134.99  
Tablet (Electronics): \$359.99  
Smartwatch (Electronics): \$269.99  
Camera (Electronics): \$494.99  
Apple (Fruit): \$0.30  
Banana (Fruit): \$0.20  
Orange (Fruit): \$0.10  
Grape (Fruit): \$0.40  
Book (Education): \$15.99  
Notebook (Education): \$4.99  
Pen (Education): \$1.99  
T-Shirt (Clothing): \$19.99  
Jeans (Clothing): \$49.99  
Sneakers (Clothing): \$79.99

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Final list of products:

Laptop (Electronics): \$999.99  
Smartphone (Electronics): \$699.99  
Headphones (Electronics): \$149.99  
Tablet (Electronics): \$399.99  
Apple (Fruit): \$0.30  
Banana (Fruit): \$0.20  
Book (Education): \$15.99  
T-Shirt (Clothing): \$19.99  
Camera (Electronics): \$549.99  
Smartwatch (Electronics): \$299.99  
Orange (Fruit): \$0.10  
Grape (Fruit): \$0.40  
Notebook (Education): \$4.99  
Pen (Education): \$1.99  
Jeans (Clothing): \$49.99  
Sneakers (Clothing): \$79.99

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Question 3: Apply PHP string manipulation functions to process text by performing the following 3 tasks:

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Original distorted product descriptions:

**Wireless Headphones:** PREMIUM\_SOUND\_QUALITY\_WITH\_NOISE\_CANCELLATION!!!  
**Smart Watch:** fitness\_tracker\_\_heart\_rate\_monitor\_\_water\_resistant  
**Laptop Stand:** ADJUSTABLE\_ERGONOMIC\_DESIGN\_FOR\_BETTER\_POSTURE@@@###  
**USB-C Hub:** multi\_port\_adapter\_\_4K\_HDMI\_\_USB\_3.0\_\_SD\_CARD\_READER!!!  
**Bluetooth Speaker:** PORTABLE\_WATERPROOF\_SPEAKER\_WITH\_360\_DEGREE\_SOUND\$\$\$  
**Mechanical Keyboard:**  
RGB\_BACKLIT\_\_GAMING\_KEYBOARD\_\_MECHANICAL\_SWITCHES@@@  
**Webcam:** HD\_1080P\_WEBCAM\_WITH\_BUILT\_IN\_MICROPHONE\_AND\_AUTO\_FOCUS###  
**Mouse Pad:** LARGE\_EXTENDED\_MOUSE\_PAD\_\_ANTI\_SLIP\_BASE\_\_WATER\_RESISTANT!!!  
**Phone Case:** SHOCKPROOF\_PROTECTIVE\_CASE\_WITH\_SCREEN\_PROTECTOR\_\_\_\_  
**Cable Organizer:** CABLE\_MANAGEMENT\_SOLUTION\_\_CLIPS\_AND\_TIES\_NEAT

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### Sanitized product descriptions:

**Wireless Headphones:** premium sound quality with noise cancellation!!!

**Smart Watch:** fitness tracker heart rate monitor water resistant

**Laptop Stand:** adjustable ergonomic design for better posture

**USB-C Hub:** multi port adapter 4k hdmi usb 3.0 sd card reader!!!

**Bluetooth Speaker:** portable waterproof speaker with 360 degree sound

**Mechanical Keyboard:** rgb backlit gaming keyboard mechanical switches

**Webcam:** hd 1080p webcam with built in microphone and auto focus

**Mouse Pad:** large extended mouse pad anti slip base water resistant!!!

**Phone Case:** shockproof protective case with screen protector

**Cable Organizer:** cable management solution clips and ties neat

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Total number of characters: 59

Total number of words: 9

Search result: [Keyword found](#)

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Extracted text: Great product! Fast ...

Keyword position: 33

Updated review: Great product! Fast delivery and excellent service. Thank you for your feedback!